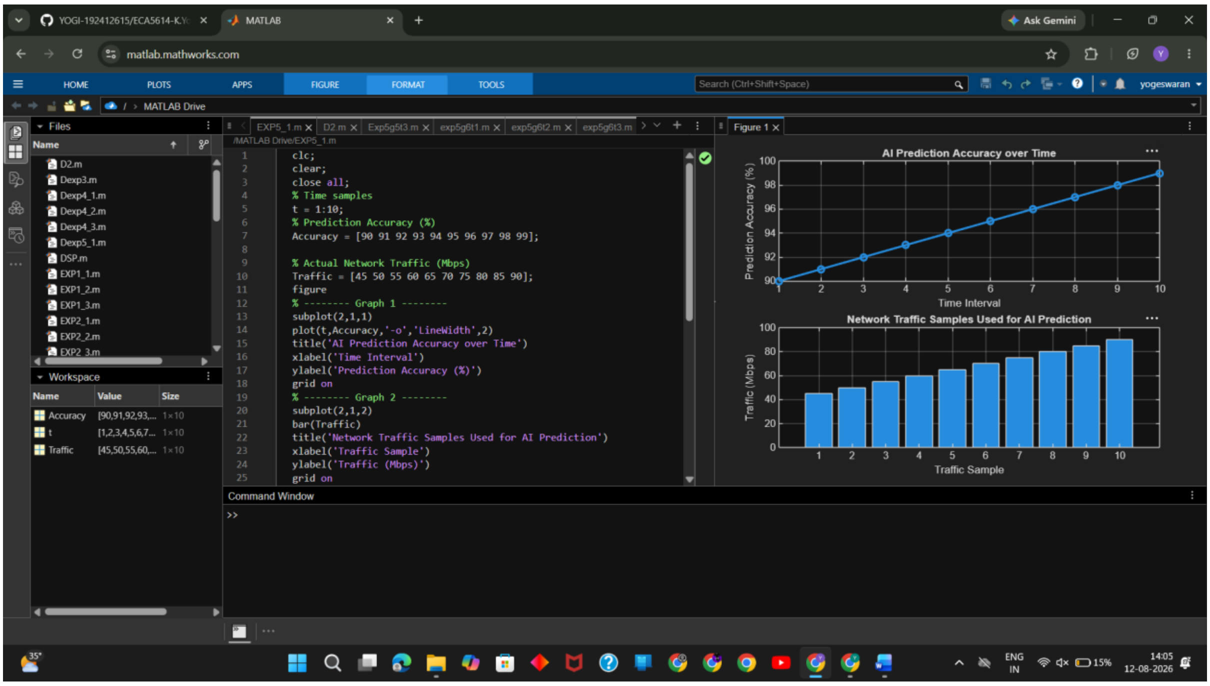
EXPERIMENT 15: AI-BASED TRAFFIC PREDICTION FOR 5G NETWORKS

Objective – 1:

MATLAB Code:

```
clc;
clear;
close all;
% Time samples
t = 1:10;
% Prediction Accuracy (%)
Accuracy = [90 91 92 93 94 95 96 97 98 99];

% Actual Network Traffic (Mbps)
Traffic = [45 50 55 60 65 70 75 80 85 90];
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(t,Accuracy,'-o','LineWidth',2)
title('AI Prediction Accuracy over Time')
xlabel('Time Interval')
ylabel('Prediction Accuracy (%)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
bar(Traffic)
title('Network Traffic Samples Used for AI Prediction')
xlabel('Traffic Sample')
ylabel('Traffic (Mbps)')
grid on
```

OUTPUT



Objective – 2: Compare Slice Utilization (%)

Matlab Code :

clc;

clear;

close all;

% Time intervals

t = 1:10;

% Actual and Predicted Traffic (Mbps)

Actual = [50 55 60 62 68 70 75 80 85 90];

Predicted = [49 54 61 63 67 71 74 81 84 89];

% Squared Error

SqError = (Actual - Predicted).^2;

% Mean Squared Error

MSE = mean(SqError);

disp(['Mean Squared Error (MSE) = ', num2str(MSE)])

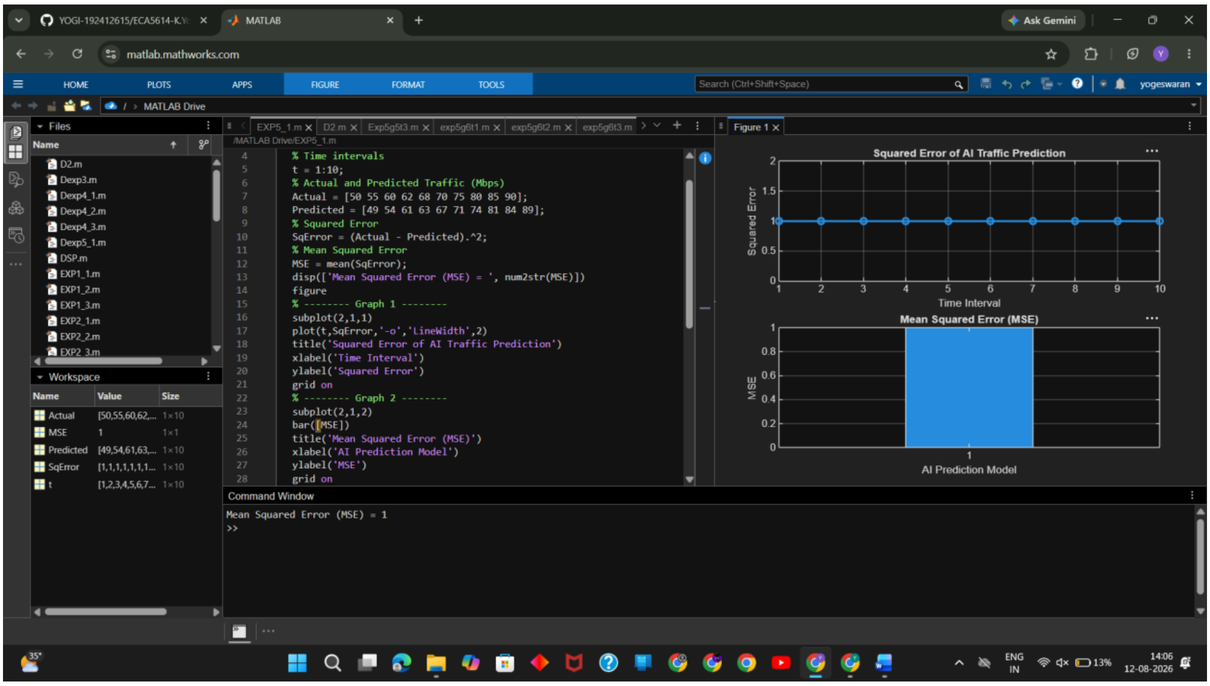
figure

% -------- Graph 1 --------

subplot(2,1,1)

```
plot(t,SqError,'-o','LineWidth',2)
title('Squared Error of AI Traffic Prediction')
xlabel('Time Interval')
ylabel('Squared Error')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
bar([MSE])
title('Mean Squared Error (MSE)')
xlabel('AI Prediction Model')
ylabel('MSE')
grid on
```

OUTPUT

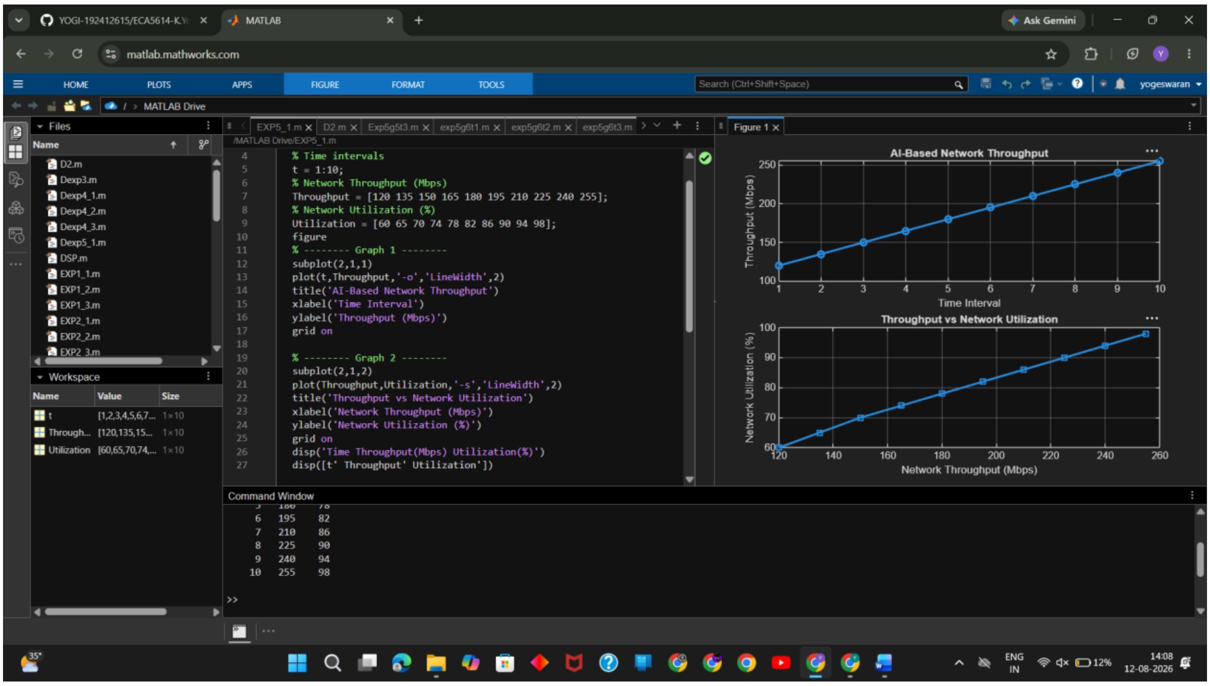


Objective – 3:

MATLAB Code:

```
clc;

clear;

close all;

% Time intervals

t = 1:10;

% Network Throughput (Mbps)

Throughput = [120 135 150 165 180 195 210 225 240 255];

% Network Utilization (%)

Utilization = [60 65 70 74 78 82 86 90 94 98];

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(t,Throughput,'-o','LineWidth',2)

title('AI-Based Network Throughput')

xlabel('Time Interval')

ylabel('Throughput (Mbps)')

grid on


% -------- Graph 2 --------

subplot(2,1,2)

plot(Throughput,Utilization,'-s','LineWidth',2)

title('Throughput vs Network Utilization')

xlabel('Network Throughput (Mbps)')

ylabel('Network Utilization (%)')

grid on

disp('Time Throughput(Mbps) Utilization(%)')

disp([t' Throughput' Utilization'])
```

OUTPUT


% Time intervals
t = 1:10;
% Network Throughput (Mbps)
Throughput = [120 135 150 165 180 195 210 225 240 255];
% Network Utilization (%)
Utilization = [60 65 70 74 78 82 86 90 94 98];
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(t,Throughput,'-o','LineWidth',2)
title('AI-Based Network Throughput')
xlabel('Time Interval')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(Throughput,Utilization,'-s','LineWidth',2)
title('Throughput vs Network Utilization')
xlabel('Network Throughput (Mbps)')
ylabel('Network Utilization (%)')
grid on
disp('Time Throughput(Mbps) Utilization(%)')
disp([t' Throughput' Utilization'])
AI-Based Network Throughput
Throughput (Mbps)
Time Interval
Throughput vs Network Utilization
Network Utilization (%)
Network Throughput (Mbps)
Command Window
6 195 82
7 210 86
8 225 90
9 240 94
10 255 98